

No Detected Interaction Between Deployed Quantization Tier and Lossy Context Compression in Extractive QA

A Corrective Paired Grid on a 12 GiB Consumer GPU

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Abstract

Local language-model deployments combine quantized weights with compressed prompts, but the accuracy cost of that combination need not be additive. We tested whether compression costs more at a smaller deployed tier using three deployments of Qwen3.8-27B on a 12 GiB laptop GPU: a partly offloaded 5.14-bpw GGUF reference, a fully resident EXL3 artifact labeled 2.0 bpw, and a fully resident 2.49-bpw GGUF. Each answered the same 117 private and 120 WildChat extractive-QA items under original context, LLMingua-2, and two stopword/code-stripping transformations, yielding 2844 scored requests. The 12 paired differences in compression cost range from -5.0 to 6.0 percentage points; none is significant after Holm correction of source-conversation-clustered tests. Every pointwise 95% interval contains zero, with an overall envelope of -11.6 to 12.9 points. These intervals do not establish equivalence. Compressor choice produces the larger observed separation: LLMingua-2 costs 15–21 points, versus 47–62 for the naive transformations at comparable character retention. Original contexts also yield more correct answers per model-serving minute within every deployment and corpus; compressor preprocessing is excluded. The practical lesson is to validate compression on the downstream task rather than rely on retention ratio alone. This corrective exploratory rerun is not an independent confirmation of the earlier, invalidated campaign. Its findings concern these deployed stacks and already-fitting inputs, not isolated weight precision or memory savings.

1 Introduction

A local deployment can lose information in two different places. Quantized weights approximate the model, while prompt compression removes or rewrites the evidence supplied at inference time. Users of a small GPU often apply both, so a useful operational question is whether the accuracy cost of compressed context grows as the deployed model artifact becomes smaller.

The answer matters for deployment. If the losses compound, a context pipeline validated with a larger artifact may fail after a switch to a smaller one. If the compression cost remains stable, weight selection and context processing can be tuned more independently. Existing quantization studies usually vary weights against unchanged inputs, while prompt-compression studies usually vary inputs against a fixed target model [7, 10, 15, 24]. Joint evaluations already exist, including GPTQ with LLMingua-2 on LongBench [1]. The narrower question here is how much the *compression cost itself* changes across practical low-bit deployments of one checkpoint, with paired uncertainty on that difference.

We evaluate that interaction with the same QA items at every point in a three-by-four grid. The three rows are real deployments of one 27B checkpoint, not laboratory-isolated bit treatments: they differ in artifact format, inference engine, and GPU residency as well as footprint. The four columns are the original context, LLMingua-2, and two deterministic rungs from a deployed stopword/code-stripping pipeline. The study runs the same grid on a public WildChat subset and a private assistant-chat corpus.

The observed compression costs change little across the three deployments: the largest descriptive spread within any corpus-condition column is 6 points. The paired interaction estimates are -5.0 to 6.0 points and none survives the familywise rule. The pointwise intervals nevertheless span -11.6 to 12.9 points across the 12 comparisons, so the result is *no detected interaction*, not demonstrated equivalence. The grid also shows a much larger separation between compressors: preserving relevant structure matters more than the fraction of characters retained in these items.

A natural reading of this design would be a “VRAM exchange rate”—how much context capacity a smaller artifact buys. This study does not measure one. All servers reserve an 8192-token context, compression is measured in characters rather than tokens, and KV-cache bytes are not recorded; VRAM savings are unmeasured. The study measures accuracy and request time after lossy input transformations across deployed tiers. That narrower question is both reproducible and directly supported by the frozen records. The evidence is a corrective exploratory rerun, not an independent confirmation: an earlier grid failed server-identity verification, and the replacement protocol was designed after its outputs had been seen. Section 3.1 explains the correction.

2 Related work

Prompt compression. LLMingua, LongLLMingua, LLMingua-2, and Selective Context remove or prioritize prompt content to reduce input length [7, 8, 10, 15]. Broader evaluations show that quality and response-length effects depend on the target model, task, and context length [24]. Kummer et al. separate compression overhead from inference and find end-to-end speedups only in matched operating regions; outside them, preprocessing can erase the serving gain [9]. Our timing measurement excludes compressor preprocessing and therefore addresses a different, narrower quantity.

Weight quantization. GPTQ and AWQ exemplify post-training weight quantization [4, 11]. Deployed local formats add engine-specific kernels and placement decisions: this study uses llama.cpp GGUF deployments [5] and an EXL3 deployment served through ExLlamaV3 and tabbyAPI [19, 21]. Those implementation choices are part of each tested tier. Quantization losses can also grow with input length and vary by method, model, and task [13].

Joint effects of weights and inputs. Ahmed et al. combine 4-bit GPTQ weights with LLMingua-2 on LongBench, comparing FP16, prompt compression, weight quantization, and their combination [1]. Their study establishes that the combination and its task-dependent trade-offs predate this work. Our contribution is narrower: paired estimates of the change in compression cost across three deployed low-bit stacks, with source-conversation-clustered uncertainty. Their LongBench F1 scores and our exact-match scores on generated questions are not directly comparable.

Ayadi et al. evaluate quantized-model robustness to word- and character-level perturbations, including filler-word deletion intended to preserve meaning [2]. That deletion overlaps with our stopword condition, but their perturbation design is not a retained-context grid comparing LLMingua-2 with aggressive code/stopword removal. Shahrabi-Farahani and Rahmati find that 4-bit bitsandbytes weights amplify proactive interference as semantically similar overwritten values accumulate [18]; their intervention adds conflicting history rather than removing context. These results show why an input-precision interaction cannot be assumed absent in general. Our search through 2026-

09-28 located no exact counterpart to the present paired deployment comparison, but its value is the bounded measurement, not priority for combining quantization and prompt compression.

3 Methods

3.1 Corrective protocol and provenance chain

The first full-grid campaign, P2G, was quarantined during release audit: its launcher had accepted any healthy HTTP endpoint on the expected port, and a stale server from an earlier session had in fact answered every request while the intended servers failed to start. No cross-tier number from that campaign is used anywhere in this paper.

P2R1 is the corrective rerun. Its protocol — deployments, question quotas, prompts, decode and scoring rules, statistical analysis, and stage order — was frozen in a machine-readable contract and committed before any inference, together with the recovery harness; every question and run contract embeds that harness’s source hashes. Because the hypotheses, corpus frames, and contaminated P2G outputs had already been observed, the protocol labels the study a corrective exploratory rerun; it is not an independent confirmation.

The analyzed run is the fifth registered attempt. The four earlier attempts stopped because of completion caps, an operator interruption, or a conservative EXL3 placement check. Each restart followed a committed operational amendment and regenerated the questions under the same seeds. Inputs, quotas, prompts, decoding, scoring, and analysis were unchanged. Earlier partial outputs are excluded; the artifact’s design record retains the attempt-level history.

Before each launch, the harness requires a cool, near-idle host with sufficient free memory. It binds the server to its process, configuration, engine revision, model identifier, and artifact hashes, then revalidates that identity for each request session. The llama.cpp loader must report the prescribed GPU placement before evaluation. Hash-chained request journals and pre/post-run artifact and CUDA-runtime checks bind the outputs to those deployments. The claims build consumes only evidence admitted by this verified record.

3.2 Hardware and deployed tiers

All requests ran on one laptop with an NVIDIA GeForce RTX 4080 Laptop GPU (12,282 MiB VRAM, 80 W power limit), 62 GiB system RAM, and an Intel Core i9-14900HX. A host inventory receipt binds the campaign to this one machine; the promotion and claims rebuild require agreement with it. The GGUF tiers used llama.cpp; the EXL3 tier used ExLlamaV3 through tabbyAPI. Each server exposed the same OpenAI-compatible interface with an 8192-token context allocation and an 8-bit quantized KV cache.

Table 1 reports three deployments of the same Qwen3.8-27B checkpoint [17]. The two GGUF artifacts come from one immutable Unsloth repository revision [22]; the EXL3 shards come from one immutable turboderp repository revision [20]. Effective file bpw is computed only for GGUF artifacts, whose header supplies the stored tensor-shape parameter count. EXL3 uses its upstream 2.00-bpw bulk label; a simple shard-bytes ratio would mix higher-precision components into a quantity that the label does not claim to represent.

3.3 Corpora and frozen evaluation sample

The public source is WildChat-1M [25], rebuilt from one pinned dataset revision with checked shard hashes. A seeded builder scanned English conversations with at least three turns and created short (target 4,000 characters) and long (target 12,000 characters) windows, with and without code, giving a frame of 160 WildChat windows (40 per stratum). The private source is the author’s assistant-chat corpus (TSI), frozen as a frame of 126 windows (40 short/no-code, 40 short/code, 6 long/no-code, and 40 long/code; the long/no-code stratum is exhausted by its source material).

Tier	Artifact	Size	bpw	Engine	Placement
higher footprint	UD-Q4_K_XL	16.4 GiB	5.14 effective	llama.cpp	partial, 33/66
low, EXL3	SC_2.00bpw_H3	9.5 GiB	2.0 label	ExLlamaV3	full GPU
low, GGUF	UD-IQ2_S	7.8 GiB	2.49 effective	llama.cpp	full GPU, 65/65

Table 1: Tested artifact–engine–placement configurations. Size, format, engine, and placement change together; “tier” is shorthand for the complete deployment, not a controlled weight-bit intervention. The placement fractions are the loader’s own logged offload readbacks.

Corpus	QA items	source windows	short/no code	short/code	long/no code	long/code
WildChat	120	120	30	30	30	30
private	117	117	37	37	6	37

Table 2: Frozen stratified evaluation sample. One question per source window; interaction uncertainty is clustered by source conversation (120 public and 107 private clusters).

Private text, questions, answers, and per-item outputs remain local; only aggregates enter this paper.

Unlike the quarantined campaign, P2R1 fixes the analyzed sample by stratified quota before any evaluation. Question selection kept exactly one accepted question per source window and filled predeclared per-stratum quotas: 117 TSI items (37/37/6/37 across the four strata) and 120 WildChat items (30/30/30/30). Table 2 records the frozen sample; each item is answered by all three deployments under all four conditions.

3.4 Context transformations

Each source window has four versions. O is the frozen original. A removes stopwords, retaining lowercased alphabetic tokens of at least three characters. E first strips code blocks, error output, and inline code with deterministic regular expressions, then applies A. Both are recomputed for P2R1 from the frozen originals by one committed deterministic implementation. L2 uses the `microsoft/llmlingua-2-xlm-roberta-large-meetingbank` checkpoint on CPU under a pinned toolchain; its requested rate is matched per window to E’s character-retention rate. No derived text from the quarantined campaign enters P2R1. Table 3 summarizes the four conditions. Every generation and evaluation prompt embeds at most the first 14,000 characters of its (transformed) window, a cap frozen in the protocol; the reported retention percentages are computed over whole windows.

3.5 Question generation, scoring, and timing

Only the identity-verified higher-footprint deployment generated questions. For every frame window it proposed two candidate factual questions per attempt, each with a short verbatim answer, under a deterministic seeded request schedule: each window had at most three generation attempts at predeclared retry seed offsets of 0, 1000000, and 2000000, and a generator request error advanced only to the next predeclared retry seed. A deterministic verifier kept a candidate pair only when the answer occurred verbatim in the original window text and its normalized form occurred in the normalized text, within the window’s first 14,000 characters (2–80 answer characters, at most eight words; questions 13–500 characters). Selection then kept at most one accepted question per window, filling the exact stratified quotas of §3.3 by a deterministic hash ranking that prefers distinct source conversations. The generation journal privately retains the exact response bytes and a separately hashed parser-visible projection, so acceptance decisions can be replayed without

Condition	Transformation	Characters kept (public/private)	Role
O	none	100/100 %	anchor
L2	LLMLingua-2, requested rate matched to E	56/58 %	learned
A	deterministic stopword filter	64/65 %	naive
E	code strip plus stopword filter	51/53 %	naive

Table 3: Lossy context conditions. Retention is measured from aggregate characters, not tokenizer tokens, and is not a KV-cache measurement.

republishing conversation content. The question server was then shut down, the host had to pass the same cool-state gate, and evaluation reloaded each deployment from a cold start with one recorded, unmeasured warmup request.

Every evaluation request showed one version of a window and one question, then required a final **ANSWER: <short exact phrase>** line. Primary correctness is exact match after case folding, whitespace collapsing, and trimming of leading and trailing quote, backtick, and common punctuation characters. A containment-based lenient score is secondary. Requests used temperature 0.2, top-p 0.95, a 2048-token completion cap, and a 600-second timeout. The protocol specified one seeded decode per cell, and the harness requested seed 42 for every deployment. The llama.cpp endpoints accept that field, but the pinned TabbyAPI revision 4a4f9f4 has no per-request seed parameter and ignores it. For EXL3, the recorded seed therefore describes the request, not an applied random state; replaying it does not make the sampled response reproducible. Separately, the four conditions for each item ran back-to-back in a seeded shuffled order within each deployment. An HTTP or model error recorded under a revalidated endpoint identity is retained once as an incorrect outcome with its transport evidence; it is never silently reissued. The frozen ledgers record 0 evaluation request errors across the 2844 scored requests and 0 question-generation request errors across the generation attempts.

Reported wall time is the end-to-end chat-completion HTTP call only: it includes server prefill and generation, and it excludes identity-guard checks, model load, cool-state gate waits, warmups, and the offline compressor preprocessing, which ran before any timed request. The frozen llama.cpp logging verbosity adds bounded runtime overhead that is included in llama.cpp wall times and not assumed to be zero. The private rows retain exact response bytes and, for the two llama.cpp tiers, server-reported usage telemetry; tabbyAPI returned no usage object for these chat completions, so the EXL3 rows carry none. The analysis does not use token counts, and no per-token timing series or prefill/decode timing split exists, so request time is reported descriptively and no timing mechanism is claimed.

3.6 Statistics

Within a deployment, O and each compressed condition share items. We report the paired accuracy cost $O - C$, its Newcombe method-10 95% interval [14], and an exact two-sided McNemar test [12]. Single-condition accuracy uses Wilson intervals [23].

For cross-tier interaction, let $Y_{itc} \in \{0, 1\}$ be exact correctness for item i , tier t , and condition c . For a low tier t and compression condition C , define

$$z_i = (Y_{itO} - Y_{itC}) - (Y_{iHO} - Y_{iHC}), \quad \hat{\theta} = \frac{1}{n} \sum_i z_i,$$

where H is the higher-footprint deployment. Positive $\hat{\theta}$ means compression cost more accuracy at the low tier. P2R1 uses an intercept-only CR2 source-conversation-cluster-robust variance estimator under an iid working model and Satterthwaite reference degrees of freedom [3, 16]. Source

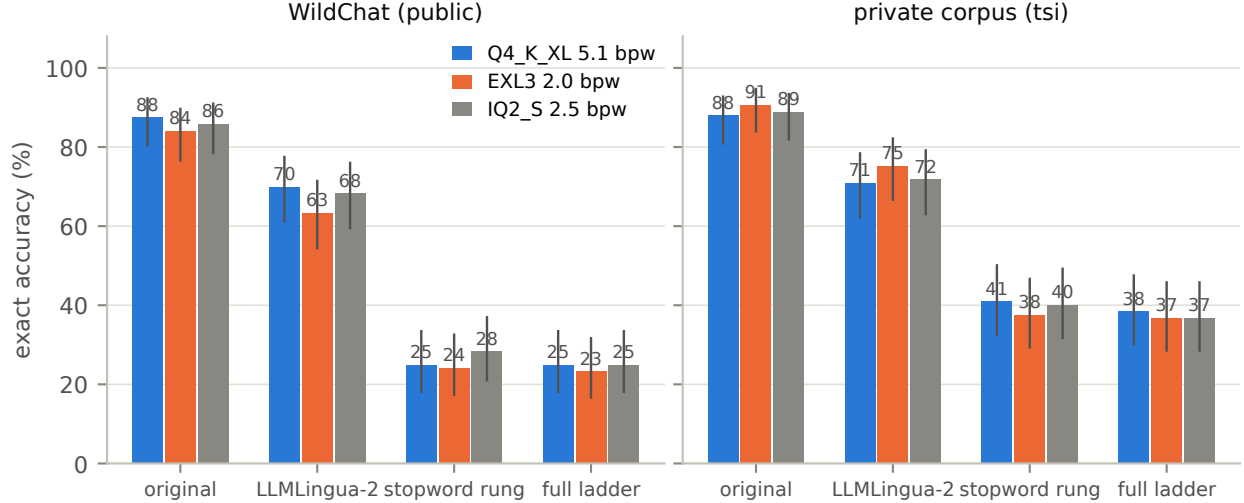


Figure 1: Exact accuracy in all 24 cells, with 95% Wilson intervals. In this and the following figures, “stopword rung” is condition A and “full ladder” is condition E. Similar bar heights within a condition are descriptive; overlapping marginal intervals are not a test of equality.

conversations are treated as independent clusters; dependence within a conversation remains unrestricted. All 12 effective degrees of freedom must be at least five within numerical tolerance, or the analysis halts for statistical review before any result branch is set.

The primary study-level branch is set only by Holm’s step-down procedure applied to the 12 model-assisted approximate CR2/Satterthwaite two-sided p -values at familywise level 0.05 [6]: the study reports a detected interaction if and only if Holm rejects at least one contrast. The 95% clustered intervals are pointwise and descriptive; they do not set the branch. The exhaustive cluster sign-flip calculation is secondary, and exact only conditional on symmetric, exchangeable source-cluster contrast signs under a sharp zero-effect null. The inference conditions on one frozen Q4-generated, eligibility-selected question set and one sampled decode per cell; it does not integrate alternate question-generation seeds or run-to-run variation and is not design-based inference for the deterministic stratified frame. Failure to reject is not evidence of equivalence: no smallest effect of interest, equivalence margin, equivalence test, or power target was pre-specified.

4 Results

4.1 Accuracy across the grid

Figure 1 and Table 4 report exact accuracy with 95% Wilson intervals for all 24 cells; original-condition point estimates span 84–91%. The within-condition similarity of the three deployments is descriptive; the formal cross-tier comparison is the paired analysis of §4.2.

4.2 No cross-tier interaction was detected

Table 5 reports the paired cross-tier estimand. Estimates range from -5.0 to 6.0 points, no contrast is rejected by the Holm familywise rule, and all pointwise 95% intervals include zero. Their full envelope is -11.6 to 12.9 points. This envelope summarizes the individual intervals; it is neither a simultaneous confidence bound nor a minimum detectable effect. The data show similar observed compression costs but leave practically important interactions unresolved. They do not establish independence or equivalence.

Corpus	Deployment	O	L2	A	E
WildChat ($n = 120$)	Q4_K_XL	88 [80,92]	70 [61,77]	25 [18,33]	25 [18,33]
	EXL3 2.0	84 [77,90]	63 [54,71]	24 [17,33]	23 [17,32]
	IQ2_S	86 [78,91]	68 [60,76]	28 [21,37]	25 [18,33]
private ($n = 117$)	Q4_K_XL	88 [81,93]	71 [62,78]	41 [33,50]	38 [30,48]
	EXL3 2.0	91 [84,95]	75 [67,82]	38 [29,47]	37 [29,46]
	IQ2_S	89 [82,93]	72 [63,79]	40 [32,49]	37 [29,46]

Table 4: Exact accuracy (%) and 95% Wilson intervals. Original-condition point estimates range from 84% to 91%.

Corpus	Condition	Low tier	Interaction pp [95% CI]	CR2 p	Holm p
WildChat	L2	EXL3	3.3 [−3.3, 9.9]	0.319	1.000
	L2	IQ2_S	0.0 [−7.0, 7.0]	1.000	1.000
	A	EXL3	−2.5 [−8.0, 3.0]	0.368	1.000
	A	IQ2_S	−5.0 [−11.6, 1.6]	0.134	1.000
	E	EXL3	−1.7 [−7.4, 4.1]	0.566	1.000
	E	IQ2_S	−1.7 [−8.3, 5.0]	0.619	1.000
private	L2	EXL3	−1.7 [−8.1, 4.7]	0.596	1.000
	L2	IQ2_S	0.0 [−5.9, 5.9]	1.000	1.000
	A	EXL3	6.0 [−0.9, 12.9]	0.089	0.978
	A	IQ2_S	1.7 [−4.7, 8.1]	0.595	1.000
	E	EXL3	4.3 [−0.1, 8.6]	0.055	0.664
	E	IQ2_S	2.6 [−3.1, 8.2]	0.368	1.000

Table 5: Marginal paired difference-in-differences against the higher-footprint deployment. Positive values mean the compressed condition cost more at the low tier. Intervals use a source-conversation-clustered CR2 variance estimate and are pointwise; p is the two-sided Satterthwaite-reference value and Holm p its step-down adjustment. The exact cluster-sign-symmetry calculation is secondary and is not shown. These are not equivalence tests.

4.3 Compressor choice is the dominant observed effect

Table 6 reports the paired within-deployment costs, and Figure 2 plots each cost against character retention. LLMingua-2 retains 56% of characters in the public arm (58% private) and costs 15–21 points. The deterministic rungs retain 51–64% publicly (53–65% private) and cost 47–62 points. Retention fraction alone therefore does not predict QA damage in this grid. The strongest comparison is L2 against E: their requested per-window rates are matched, yet their accuracy costs differ by tens of points. This supports evaluating the compressor on the downstream task rather than treating character ratio or embedding similarity as a sufficient quality measure.

4.4 Serving throughput after compression

Figure 3 plots correct answers per model-serving minute. Original contexts produce 1.1–5.5 exact-correct answers per model-serving minute across the six deployment–corpus pairs. LLMingua-2 produces 0.4–3.4, and the naive conditions produce 0.1–1.0. Mean request time is 9.7–48.7 seconds for O and 22.5–180.1 seconds for the naive conditions; per-cell means for every condition, including L2, are in the released claims file. The analysis does not use the stored usage telemetry, and no per-token timing series exists, so whether the difference comes from prefill, decode length, decode rate, or another server behavior is left unresolved; the measurement also omits the offline compression pass. The supported statement is therefore limited: for these already-fitting inputs, within each

Corpus	Deployment	Δ L2 [CI]	Δ A [CI]	Δ E [CI]
WildChat	Q4_K_XL	18 [8,27]	62 [52,70]	62 [52,70]
	EXL3 2.0	21 [11,30]	60 [50,68]	61 [51,69]
	IQ2_S	17 [8,27]	57 [47,66]	61 [50,69]
private	Q4_K_XL	17 [8,26]	47 [36,57]	50 [38,59]
	EXL3 2.0	15 [7,24]	53 [42,62]	54 [42,63]
	IQ2_S	17 [8,26]	49 [37,58]	52 [40,62]

Table 6: Paired percentage points lost relative to original context, with 95% Newcombe method-10 intervals. The per-cell exact McNemar p -values are computed in the released claims file rather than tabulated here; they establish damage relative to O, not equality across tiers.

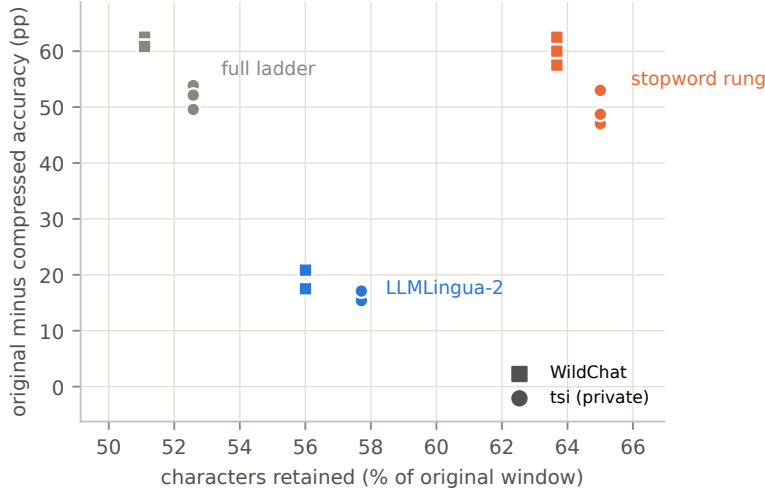


Figure 2: Paired accuracy cost against aggregate character retention. Six points per condition represent three tiers in each corpus.

of the six deployment–corpus pairs, every compressed condition yields fewer correct answers per recorded serving minute than the original context, and its mean request time is longer. The pooled ranges above overlap only because the higher-footprint deployment is slower than either low tier under every condition. It does not imply that compression cannot improve latency for longer prompts or other hardware; controlled studies find such gains in matched operating windows [9].

4.5 Operational answer survival

Figure 4 conditions accuracy on the survival tag. The survival covariate marks a gold answer as present when its normalized form (case-folded, whitespace-collapsed, with quote, backtick, bracket, and punctuation edges trimmed) occurs in the identically normalized compressed text. Under that operational rule, applied to the L2 and E conditions (the covariate is not computed for A), 198 unique item–condition pairs are tagged deleted. Each is evaluated at all three tiers, producing 594 tier-level evaluations, of which 10 are exact-correct. This is an observed count, not proof about what reconstruction is possible.

LLMLingua-2 retains the operational gold in 96/120 public items (80%; private 84%), and exact accuracy among survivors is 79–90%. E retains the gold in 30% of public items (private 39%), with survivor accuracy of 78–93%. Answer availability and the structure needed to locate an answer are therefore both part of compression damage in this grid.

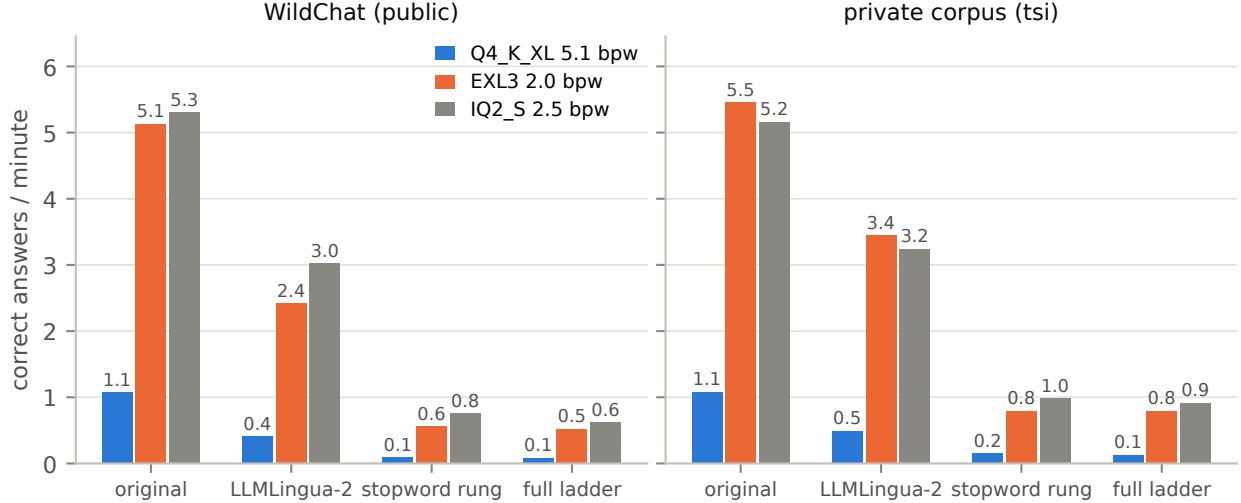


Figure 3: Correct answers per model-serving minute. Compressor preprocessing is excluded. The figure is descriptive and does not identify the timing mechanism.

4.6 Public and private arms

Both arms show the same descriptive ordering of conditions at every deployment, with O above L2 and L2 above both naive rungs, and neither arm contributes a contrast that the Holm rule rejects. The L2 cell accuracies span 63–75% across both arms. This agreement is useful robustness evidence, but it is not a formal equivalence or replication test. Every central direction is visible in the public arm; the private arm adds only aggregate evidence from one motivating workload.

5 Discussion

The principal result is modest but operationally useful. Within one extractive-QA design, switching among the three deployed tiers did not reveal a larger penalty for compressed context. The interaction point estimates are small relative to the condition effects reported in §4.3. For a practitioner using these exact stacks, the evidence favors validating the compressor over choosing a different compressor for each tier.

The uncertainty limits that recommendation. The pointwise intervals span −11.6 to 12.9 points across the comparisons; none defines a study-wide detection threshold. An operator who needs to rule out a consequential extra loss should pre-specify an acceptable margin and power an equivalence or non-inferiority study, ideally on tasks with meaningfully different clean-input performance across tiers.

The artifact–engine framing also matters. The lower-footprint tiers are fully GPU resident; the higher-footprint GGUF is partially offloaded. The study asks whether compression cost is robust across those useful deployment choices. It does not support a weight-precision causal claim: no single component of the deployment stack receives attribution.

Finally, compression is not interchangeable with capacity in this evidence. Character retention is only a rough input-size measure, all requests already fit, and the context allocation is fixed. The grid measures the cost of discarding input that could have been kept. Whether compression is beneficial when it makes an otherwise impossible request fit is a different experiment.

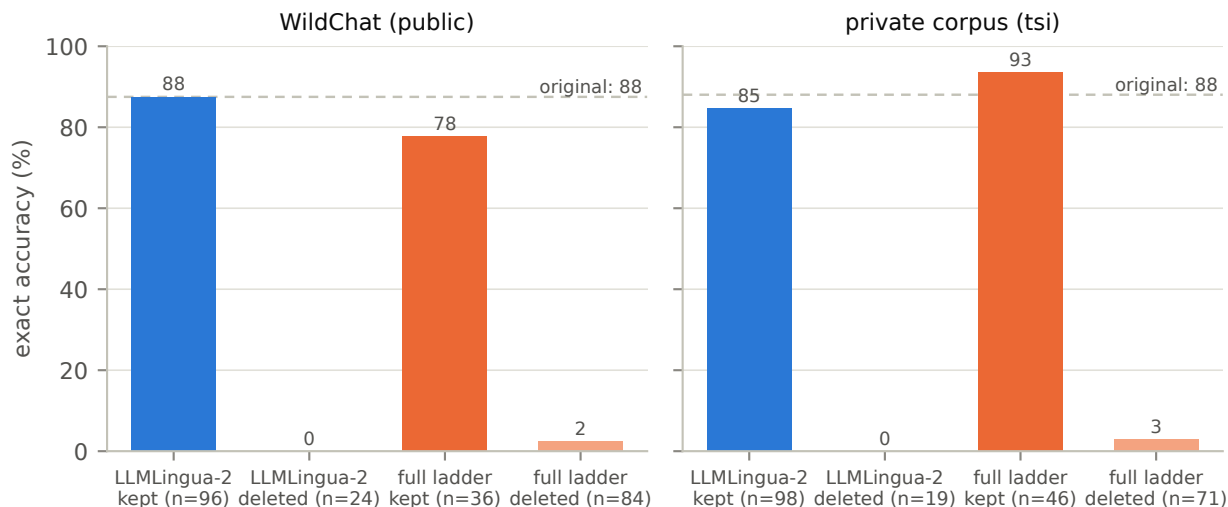


Figure 4: Exact accuracy conditional on the operational answer-survival tag for the higher-footprint deployment. “Deleted” means the normalized gold no longer occurs in the identically normalized compressed text under the frozen survival rule.

6 Limitations

The corrective chronology is the first limitation: P2R1 was designed after the hypotheses, corpus frames, and contaminated outputs of the quarantined campaign had been observed. Fresh seeded questions and an identity-verified harness repair the measurement chain, but the study remains a corrected measurement rather than a replication, and its inferences condition on one frozen question set and one sampled decode per cell. The EXL3 server did not apply the requested evaluation seed, and the study does not estimate decode-to-decode variability.

The evidence covers one dense 27B checkpoint, one laptop, three deployed artifact-engine-placement configurations, and short-answer extractive QA. Reasoning, synthesis, tool use, and generation-heavy tasks may show a different interaction. Because engine, format, effective precision, footprint, and GPU residency co-vary, no component receives a causal attribution.

The interaction intervals are pointwise and remain wide; no equivalence margin or power target was specified. Original-condition accuracy is high (84–91%), which reduces room to observe compression helping. Questions were generated by the higher-footprint deployment, creating possible selection advantage for that tier. The private long/no-code stratum contains only 6 windows, so long-context claims rest mostly on the other strata. Timing is observational within item-interleaved server sessions and supports serving-time descriptions, not a mechanism. Character retention does not measure token retention or KV memory. Prompt injection in source conversations was not separately scored and could affect the exact-answer protocol.

The private corpus cannot be released. WildChat is available under ODC-By, but that database license does not necessarily clear independent rights in every conversation, so raw windows, excerpts, questions, answers, and responses stay private in both arms and only aggregate statistics are released.

7 Reproducibility and data availability

All reported numbers derive from a single promotion record binding the frozen outputs to the protocol, harness hashes, server-identity and placement checks, host inventory, and artifact/CUDA-runtime receipts. Rebuilding claims also verifies that chain. Exact model artifacts are identified by

upstream repository, immutable revision, filename, byte count, SHA-256, and license.

Question generation uses the separate llama.cpp Q4 endpoint with the fixed seed schedule above; source selection and item ordering are also seeded. These controls do not make the EXL3 evaluations reproducible from seed 42. Analysis reproduction instead uses the exact saved outputs. The generated questions and gold answers are derivatives of conversation content: releasing them requires a separate rights and privacy decision, so the public package contains aggregate statistics, builder and analysis code, environment records, and provenance receipts rather than raw text. The accompanying aggregate artifact, version 1.0.0, is identified by [doi:10.5281/zenodo.22847291](https://doi.org/10.5281/zenodo.22847291). For analysis-only reproduction, run `python3 check_claims.py` at the artifact root. The standard-library verifier recomputes the reported intervals, tests, Holm decisions, and manuscript numbers from aggregate cluster histograms; it needs no model, GPU, or network access. Full inference replication is a separate task: `FULL_RERUN.md` specifies the pinned artifacts, environment, and harness commands. External reruns must regenerate public questions and cannot reproduce the unreleased private corpus.

8 Ethics and disclosure

WildChat’s maintainers report de-identification and removal of identified sensitive conversations, but residual sensitive material remains possible in real-world chat logs. No conversation text is quoted in the manuscript or included in the public release. Private assistant-chat material is reported only as aggregates and was scanned against release text.

AI assistance. Claude (Anthropic) and Codex (OpenAI) assisted with evaluation-harness development, experiment orchestration, evidence auditing, analysis and figure code, citation checking, and manuscript drafting and revision. Matthew Schwartz directed the research and is responsible for its methods, results, interpretation, claims, citations, rights, and released artifacts.

9 Conclusion

Across three real deployments of one 27B checkpoint, this corrective exploratory rerun detected no cross-tier accuracy interaction between quantization tier and lossy context compression on paired extractive QA. The estimated differences in compression cost are -5.0 to 6.0 points, no contrast survives the Holm familywise rule, and the pointwise interval envelope is -11.6 to 12.9 points. The result is therefore a bounded non-detection, not proof of independence. Within the same grid, compressor choice changes accuracy by tens of points: LLMingua-2 loses far less than a stopword/code-stripping pipeline at similar character retention. For these already-fitting inputs, the original context also yields the most correct answers per recorded serving minute on every deployment. The practical lesson is to validate context transformations on the actual downstream task and deployed stack. Claims that any extra loss is acceptably small require a separate, adequately powered equivalence design.

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